BUSINESS REQUIREMENTS DOCUMENT

PROJECT DETAILS

PROJECT NAME		
SafeSignal		
CREATOR		
Hadi Najem, Mario Koubayati, Elie Wakim, Salim Bakhos		
DOCUMENT NO.	DATE	VERSION NO.
1	12/05/2026	4.1

1. EXECUTIVE SUMMARY SNAPSHOT

This Business Requirements Document defines the current SafeSignal project requirements based on the implemented repository. The audience for this document is the project team, reviewers, instructors, and stakeholders who need a clear understanding of what the SafeSignal platform is expected to provide.

SafeSignal is a community safety reporting platform. It allows mobile users to submit structured safety incident reports, view nearby incident activity, track their own reports, manage account preferences, and respond to privacy-aware witness prompts. It also provides a web dashboard for authorized moderators, admins, and law-enforcement users to review reports, manage users, inspect operational activity, and analyze safety trends.

The project includes four main technical parts:

- A React Native mobile application for community users.
- An Express/PostgreSQL backend API for authentication, incident management, location-aware queries, notifications, role-based access, and workflow support.
- A React/Vite moderator dashboard for staff review, analytics, user management, and law-enforcement workflows.
- An ML service for classification, risk scoring, toxicity checks, duplicate support, witness synthesis, and analytics insights.

The main business driver is the need for a structured and privacy-aware reporting workflow. Informal reporting through phone calls, messages, or social media can make reports difficult to verify, prioritize, track, and analyze. SafeSignal addresses this by connecting mobile incident submission with staff review, AI-assisted analysis, dashboard analytics, and privacy-preserving witness corroboration.

SafeSignal is not a direct emergency dispatch system and is not a replacement for emergency services. The mobile app explicitly instructs users to call emergency services if they are in immediate danger.

2. PROJECT DESCRIPTION

SafeSignal supports community safety reporting from the moment a user submits an incident through review, analysis, and follow-up. A community user can register or log in, create a report, provide incident details, add a location, optionally attach photos, report anonymously, save a draft, and later track the report status.

The current process that SafeSignal improves is informal community reporting. Without a dedicated system, safety information may be shared through scattered channels, making it difficult to:

- collect consistent report details
- identify serious or urgent reports
- detect duplicate reports
- protect reporter and witness privacy
- provide report status visibility
- analyze safety trends over time
- support staff and law-enforcement workflows

SafeSignal addresses these challenges with a structured platform. Submitted reports are validated and stored by the backend. The system may use ML-assisted processing to support category classification, risk scoring, toxicity checks, duplicate review, and analytics insights. Moderators can review incoming reports through a dashboard, inspect details and timelines, apply moderation decisions, and escalate valid reports. Law-enforcement users can access role-protected operational workflows for relevant incidents.

The project also includes the Witness Constellation feature. For eligible incidents, nearby consenting users may receive a prompt asking whether they noticed related activity. Their responses are summarized into aggregate confidence states without exposing private witness identities in public or mobile views.

3. PROJECT SCOPE

The SafeSignal scope includes the mobile reporting application, backend API, moderator dashboard, analytics, law-enforcement workflow, ML-assisted processing, and Witness Constellation feature.

Project Goals

- Provide a mobile application for structured community incident reporting.
- Allow users to submit reports with title, description, category, severity, date/time, location, anonymity preference, and optional photos.
- Allow users to save local drafts and continue reports later.
- Display active and resolved incidents on the mobile map.
- Allow users to track their submitted reports.
- Provide authenticated role-based access for citizens, moderators, admins, and law-enforcement users.
- Provide moderator tools for report review, prioritization, filtering, escalation/verification, rejection, duplicate handling, timelines, and ML summary review.
- Provide admin tools for user management, role updates, suspensions, and administrative controls.
- Provide a Data Analysis Center for operational metrics and AI-supported insights.
- Provide law-enforcement workflows for operational incident handling.
- Provide privacy-aware witness corroboration through Witness Constellation.
- Protect sensitive reporter, witness, and location data.

Project Tasks

- Requirements gathering and project documentation.
- Mobile UI/UX design and React Native development.

- Backend API development using Express and PostgreSQL/PostGIS.
- Authentication, authorization, and role-management implementation.
- Incident submission, status lifecycle, map, feed, and report-history implementation.
- Moderator dashboard development.
- Data Analysis Center implementation.
- Law-enforcement workflow implementation.
- ML service integration for classification, risk, toxicity, duplicate support, synthesis, and insights.
- Witness Constellation implementation.
- Testing, debugging, validation, and final documentation.

Project Deliverables

- SafeSignal mobile application prototype.
- SafeSignal backend API.
- SafeSignal moderator/admin/law-enforcement dashboard.
- Data Analysis Center dashboard.
- ML service.
- Witness Constellation feature.
- Test cases and validation artifacts.
- Final project documentation and BRD.

Cost and Deadline Notes

The project is primarily an academic or prototype-based software solution. Expected costs are limited to development tools, API integrations, database hosting, mobile test devices or emulators, notification infrastructure if configured, and deployment resources if the prototype is hosted.

The project timeline follows the normal software development sequence: planning and requirements analysis, design and prototyping, backend/mobile/dashboard implementation, ML and analytics integration, witness corroboration implementation, testing, cleanup, documentation, and final presentation.

In-Scope Items and Out-of-Scope Items

IN-SCOPE ITEMS	OUT-OF-SCOPE ITEMS
Item 1: Mobile incident reporting application.	Item 1: Direct integration with official government emergency dispatch systems.
Item 2: Structured report form with incident details, location, anonymity, and optional photos.	Item 2: Shake-triggered emergency activation.
Item 3: User authentication, Google sign-in, profile/account access, and role-based access.	Item 3: AI-generated fake voice calls or simulated phone calls.
Item 4: Local draft saving and later continuation.	Item 4: Personal emergency-contact alerting or trusted-contact management.
Item 5: Active/resolved incident map and public feed surfaces.	Item 5: Live emergency-session tracking for personal contacts.

IN-SCOPE ITEMS	OUT-OF-SCOPE ITEMS
Item 6: Moderator report review queue and report detail workflow.	Item 6: Official police, hospital, shelter, or emergency-station routing.
Item 7: Admin user management and system controls.	Item 7: Hardware devices, wearables, or physical infrastructure.
Item 8: Law-enforcement operational workflow for escalated incidents.	Item 8: Full commercial deployment and 24/7 customer support operations.
Item 9: Data Analysis Center metrics and AI-supported insights.	Item 9: Multi-language localization beyond the current app language.
Item 10: Witness Constellation corroboration and aggregate confidence display.	Item 10: Advanced threat prediction beyond implemented ML support.

4. PROPOSED PROCESS

The proposed SafeSignal process creates a structured path from community report submission to staff review, analytics, and follow-up.

1. User Access

a. The user registers or logs in through the mobile app.

b. The user may use email/password authentication or Google sign-in.

c. The backend provides authenticated access for protected mobile and dashboard APIs.

2. Incident Report Creation

a. The user opens the mobile report form.

b. The user enters title, description, date/time, location, and optional photos.

c. The user chooses anonymity preference.

d. The user provides category/severity manually or uses ML-assisted defaults where enabled.

e. The user saves the report as a local draft or submits it.

3. Backend Processing

a. The backend validates required fields.

b. The backend stores the incident and related report metadata.

c. The backend may call ML services for classification, risk scoring, toxicity checks, duplicate review, and insight support.

d. The backend updates incident status and ML metadata according to implemented rules.

e. Staff notification events may be emitted for important incident activity.

4. Witness Constellation

a. For eligible incidents, the backend may open a Witness Constellation.

b. Nearby users who have provided location consent may receive a witness prompt.

c. A witness may submit a predefined signal and optional short note.

d. The system aggregates responses into confidence states such as single report, corroborated, mixed signals, activity not confirmed, or likely ended.

e. Public and mobile views show only approved aggregate information.

5. Moderator Review

a. Moderators and admins log in to the dashboard.

b. They view report queues with filtering, search, sorting, and priority ordering.

c. They inspect report details, location context, timelines, ML summaries, duplicate candidates, and constellation state.

d. They may escalate/verify, reject, update category, link duplicates, or continue to the next report according to role permissions.

6. Law-Enforcement Workflow

a. Authorized law-enforcement users and admins access the law-enforcement dashboard.

b. They view relevant incident queues.

c. They update operational incident statuses.

d. They close cases with supported closure outcomes.

e. Disclosure and location-fuzzing controls may be applied where supported.

7. Analytics and Administration

a. Authorized moderators and admins use the Data Analysis Center to view volume, response, SLA, heatmap, category, hotspot, reporter quality, and funnel metrics.

b. AI-supported insight sections may be generated when the ML service is available.

c. Admin users manage users, roles, suspensions, and administrative system actions.

5. FUNCTIONAL REQUIREMENTS

SafeSignal functional requirements are grouped by project capability. Each requirement is assigned a priority according to the table below.

- PRIORITY

VALUE	STATUS	DESCRIPTION
1	Immediate	The requirement is critical to the project’s success. Without it, the core project workflow is not possible.
2	High	The requirement is high priority for project success, but a limited MVP could still exist without full coverage.
3	Moderate	The requirement provides important value but is not the minimum reporting path.
4	Low	The requirement is useful but not required for project success.
5	Prospective	The requirement is outside the current project scope and may be considered for a future release.

- CATEGORIES (RC1)

ID	REQUIREMENT	PRIORITY	RAISED BY
FR-01	The system must allow users to register, log in, and use authenticated mobile features.	High	Users
FR-02	The system must support Google sign-in for mobile users.	High	Users
FR-03	The system must enforce role-based access for citizen, moderator, admin, and law_enforcement roles.	High	Developers
FR-04	Dashboard routes must require authentication.	High	Developers
FR-05	The mobile app must provide a structured incident report form.	High	Users
FR-06	The report form must capture title, description, category, severity, date/time, location, anonymity preference, and optional photo references.	High	Users
FR-07	The app must validate required report fields before submission.	High	Developers
FR-08	The app must allow users to save, restore, and clear a local draft after successful submission.	Low	Users
FR-09	The app must allow users to view submitted reports and report statuses.	High	Users
FR-10	The app must allow users to use current GPS coordinates or select a map location when creating a report.	High	Users
FR-11	The mobile map must display active incidents and support resolved/public-feed incident viewing.	High	Users

ID	REQUIREMENT	PRIORITY	RAISED BY
FR-12	The map must support category and timeframe filtering where implemented.	Moderate	Users
FR-13	The backend must support location-aware incident queries using stored coordinates.	High	Developers
FR-14	Moderators must see a prioritized queue of submitted, auto-processed, and auto-flagged reports.	Immediate	Moderators
FR-15	Moderators must be able to search, filter, and sort reports.	High	Moderators
FR-16	Moderators must be able to inspect report details, timelines, ML summaries, duplicate candidates, and constellation state.	High	Moderators
FR-17	Moderators must be able to escalate/verify eligible reports and reject invalid reports.	High	Moderators
FR-18	Moderators must be able to update categories and link duplicate reports where supported.	Moderate	Moderators
FR-19	Admins and moderators must be able to view user accounts.	High	Admins
FR-20	Admins must be able to suspend/unsuspend users and update user roles.	High	Admins
FR-21	Law-enforcement users and admins must be able to access a role-protected law-enforcement interface.	High	Law Enforcement
FR-22	Authorized law-enforcement users must be able to view incident queues, update operational statuses, set closure outcomes, and manage disclosure settings where supported.	High	Law Enforcement
FR-23	The Data Analysis Center must show incident volume, response metrics, SLA indicators, heatmaps, category trends, hotspots, reporter quality, and funnel data where available.	Moderate	Admins
FR-24	The dashboard must support 7-day, 30-day, 90-day, and 1-year analytics periods.	Moderate	Admins
FR-25	AI-supported insights may be generated from analytics payloads when the ML service is available.	Low	Admins
FR-26	The backend may use ML classification, risk scoring, toxicity checks, and duplicate analysis to support review workflows.	Moderate	Developers
FR-27	ML output must support staff review and must not replace human moderation decisions in sensitive workflows.	High	Moderators
FR-28	The backend must be able to create an active Witness Constellation for eligible incidents.	Moderate	Developers

ID	REQUIREMENT	PRIORITY	RAISED BY
FR-29	The system must notify nearby consenting users when they may provide witness corroboration.	Moderate	Users
FR-30	The witness prompt must allow predefined signal responses and an optional short note.	Moderate	Users
FR-31	The reporter must not be allowed to corroborate their own constellation.	High	Developers
FR-32	The system must aggregate witness responses into confidence and ongoing-assessment fields.	Moderate	Developers
FR-33	Public/mobile views must expose only approved aggregate constellation information.	High	Users
FR-34	Staff views may expose additional constellation metadata required for moderation.	Moderate	Moderators
FR-35	Protected mobile and dashboard actions must require authentication.	High	Developers
FR-36	Public incident reads must restrict sensitive reporter data.	High	Users
FR-37	Anonymous reporting must be supported at the incident level.	High	Users
FR-38	Location consent must be required before storing user location for witness prompt workflows.	High	Users
FR-39	Sensitive witness data must not be exposed in public/mobile views.	High	Users

6. NON-FUNCTIONAL REQUIREMENTS

ID	REQUIREMENT
NFR-01	The mobile app should keep report submission, map browsing, and report-history workflows responsive under normal network conditions.
NFR-02	The backend must validate input at API boundaries and return clear errors for invalid requests.
NFR-03	Dashboard pages must remain usable for repeated staff review work, including searchable and filterable report lists.
NFR-04	Sensitive views must be protected by authentication and role-based authorization.
NFR-05	Location-aware features must handle missing, invalid, or denied location data without crashing.

ID	REQUIREMENT
NFR-06	ML-service failures must not corrupt incident data or prevent core report creation unless required by validation rules.
NFR-07	Witness prompt and constellation synthesis failures must not block incident creation.
NFR-08	Analytics views must handle empty, loading, and error states clearly.
NFR-09	Public-facing contexts must avoid exposing private witness notes, reporter identity, or exact location data unless explicitly allowed by role and workflow.
NFR-10	The mobile app, backend, and dashboard should use consistent category, severity, and status definitions.
NFR-11	Staff workflows should fail safely when optional ML or notification features are unavailable.
NFR-12	The system should preserve user privacy through anonymous reporting, role-based responses, location consent, and disclosure controls.

7. GLOSSARY

TERM/ABBREVIATION	EXPLANATION
SafeSignal	The community safety reporting platform built in this project.
Incident	A safety-related report submitted by a mobile user.
Report	A user-facing or moderator-facing representation of an incident.
Reporter	The user who submitted an incident.
Moderator	A staff role that reviews, verifies/escalates, rejects, categorizes, and manages reports.
Admin	A staff role with user and system administration access.
Law Enforcement / LEI	A role and dashboard workflow for handling escalated or operational incidents.
Data Analysis Center / DAC	The dashboard area for operational analytics and AI-supported insights.
Witness Constellation	A privacy-aware feature that asks nearby consenting users for corroborating signals and summarizes responses.
Corroboration	A witness response associated with a constellation.
ML	Machine learning or AI-assisted processing used for classification, risk, toxicity, duplicate review, synthesis, and insights.
SLA	Service-level indicator used by analytics to measure response timing.

TERM/ABBREVIATION	EXPLANATION
Location Fuzzing	Reducing exactness of location data before disclosure.
Public Feed	A controlled feed of disclosed incidents shown to users.

8. REFERENCES

NAME	LOCATION
Project explainer	`docs/project-friendly-explainer.md`
Mobile app source	`Mobile-part/`
Backend API source	`backend/`
Moderator dashboard source	`moderator-dashboard/`
ML service source	`ml-service/`
Shared incident constants	`constants/incident.js`
Witness Constellation documentation	`docs/witness-constellation-feature.md`
Data Analysis Center plan	`docs/dac-production-plan.md`
Test cases	`testcases/`

9. APPENDIX

A. Supported Incident Categories

- Theft
- Assault
- Vandalism
- Suspicious Activity
- Traffic Incident
- Noise Complaint
- Fire
- Medical Emergency
- Hazard
- Other

B. Supported Severity Levels

SEVERITY	MEANING
Low	Minor incident, no immediate risk.
Medium	Moderate concern, needs attention.
High	Serious incident, urgent.
Critical	Emergency-level incident, immediate action needed.

C. Supported Incident Statuses

- Draft
- Submitted
- Auto Processed
- Auto Flagged
- In Review
- Verified
- Dispatched
- On Scene
- Investigating
- Police Closed
- Rejected
- Needs Info
- Published
- Resolved
- Archived
- Merged

D. Current Accuracy Notes

- SafeSignal currently supports incident reporting, moderation, analytics, law-enforcement workflows, ML assistance, and witness corroboration.
- SafeSignal currently does not implement shake-triggered alerts, AI fake voice calls, personal emergency-contact alerting, official emergency-station routing, or live emergency-session tracking.
- SafeSignal should not be described as directly integrated with official emergency response systems unless that integration is added later.